Sparse geometric representation of images using Gaussian splatting (Bachelor thesis)

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December 01, 2025

Abstract

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri.

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1 Introduction

In the last few years, AI and machine learning have become increasingly important. We have seen ML applied across several domains, mainly text, image and video, and audio. For image recognition, convolutional neural networks (CNN) have been quite successful. CNNs use convolutional kernels to group pixels together, such that for a 100x100, we do not need 100,000 weights to connect each pixel. These networks are designed to extract information from images, such as classification. This has become the defacto standard for image recognition in the past 10 years. (source)

Recently, another model has become prevalent: the transformer (Vaswani et al. 2023). Transformer take tokens as inputs, in most cases, these tokens are words. The transformer can predict new tokens, leading to generative models. It was discovered (an image is worth 16x16 words) that you can serialize an image into patches of 16x16 pixels and use those as tokens to train transformers successfully. This already outperforms "classical" CNNs.

For transformers in natural language, tokens have meaningful relations to each other. A classical example is Vector("King")-Vector("Man") $\cong$ Vector("Queen")-Vector("Woman) (Mikolov et al. 2013). This is less true for 16x16 image patches, these tokens are chosen with little geometric meaning in the image. This is also described in (an image is worth 16x16 words), where we can see that attention of a token is mostly on tokens that are on the same row/column axis.

Ideally, we can find a set of tokens that describe an image, similar to how a set of words describes a sentence or text. While text tokens can encode grammatical or contextual information, image tokens should capture geometric information.

One way to generate such tokens is Gaussian splatting. This has been widely used for 3D scene reconstruction. By taking multiple pictures and spawning Gaussian blobs in 3D to match each view, resulting in a 3D scene represented by Gaussians. This is done using gradient descent to train the scenes to iteratively better match each image, using L1 and SSIM loss. Recently, Gaussian splatting has also been used in 2D for image representation (Zhang et al. 2025a). This works essentially by only taking only one image, and training for that single image. This process works a lot simpler as quite a few complication for 3D do not have to be made anymore. Besides encoding tokens for transformers, 2D Gaussian splatting can also be used in scenarios where encoding is done once and decoding is done multiple times, such as textures for video games.

Tokenization of Gaussian parameters has already been studied (Dong et al. 2025). In this thesis we aim to provide more geometric value to these tokens by using Lie theory. Furthermore, we want to encode higher-frequency data using derivatives of Gaussians. Convolutional neural networks have been shown to extract information from high-frequency components; explicitly encoding these could give transformers a similar capability. (Source?)

2 Theory

We are expanding the Gaussian splatting method with an explicit geometric description in terms of a Lie group, as well as using higher order derivatives to more efficiently encode high frequency features. This is especially used in areas of high contrast, such as edges and lines.

We will spawn n initial Gaussians, each of which will represent one token, and each token will have p (tbd) parameters that describe the color values of the Gaussian layers and the geometric properties such as location, rotation, and scale.

2.1 Gaussian layers

Instead of using just the Gaussian function as is usual in Gaussian splatting, we will also use the first and second order derivatives of the Gaussian function. As defined as

$$\phi_0(x) := e^{-\|x\|^2}$$
$$\hat{\phi}_1 = \frac{\mathrm{d}}{\mathrm{d}x_1}\phi_0$$
$$\hat{\phi}_2 = \frac{\mathrm{d}}{\mathrm{d}x_1}\phi_1$$

Please note that the choice of variable (x_1 or x_2) for the derivative does not matter, as the axes are orthogonal and the function can be rotated by 90 deg to get the other derivative.

This will give us more complex features to work with. These three Gaussians will be layered in the same location. Below, we can see the derivatives when they are unscaled.

> **Todo**
>
> Add color scale

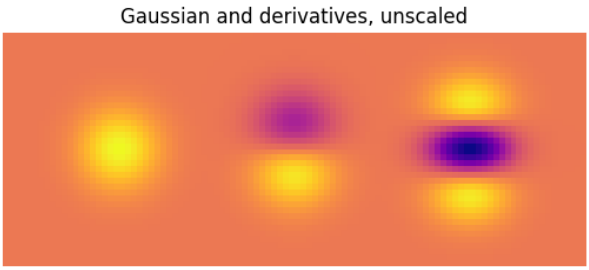


Figure 1: Unscaled (f.l.r): Gaussian (ϕ_0), first-order derivative ($\hat{\phi}_1$) and second-order derivative ($\hat{\phi}_2$) of a Gaussian.

The idea is that the "zeroth-order" derivative is used as the base color, and that the derivatives are used as accents. The first-order derivative contains a fast drop-off, which could be used for creating a sharp edge, and the second-order derivative contains a sharp bump in the middle; this could be used as lines.

Layering these as is will not result in a good overlap, however, because the "support" of these functions differs a lot. We can scale the derivatives such that their "support" matches better with the Gaussian. Scaling the first-order by 0.7 and the second-order by 0.5, we get the following Gaussians, which will be layered to create more complex structures.

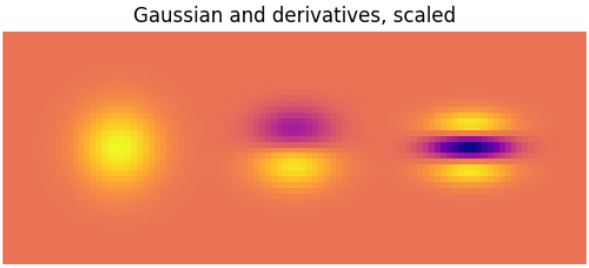


Figure 2: Scaled: Gaussian (ϕ_0), first-order derivative (ϕ_1), and second-order derivative (ϕ_2) of a Gaussian.

Each of these Gaussians has an independent set of color values. $c^{i,j}$, i for each the Gaussian layers, and j for the color channels, totalling 9 values. Each color value is not directly the color that the Gaussian attains, but rather an addition or subtraction in a certain direction for the final color. For each pixel color $pc \in [0,1]^3$, we start with a default of 0.5. This has the advantage that the image is color invertible by multiplying all the color values by -1, but also does not introduce an implicit bias to the color black, as opposed to white (or vice versa). Therefore, the color values of the Gaussians can attain any value in $\mathbb{R}$, however, most likely they will be pushed between $[-0.5, 0.5]^3$ as the input image contains only values between $[0,1]^3$. This will be elaborated on later.

2.2 Lie groups

This component for position, rotation, and scale should ideally be a Lie group. As this means our representation can benefit from all the structure and tools that Lie groups provide.

2.2.1 Affine group

Such a group G is defined as follows.

Definition 2.2.1 (The affine group with positive determinants) The *2-dimensional, real affine group with positive determinants* is defined as

$$G = \mathrm{Aff}^+(\mathbb{R}^2) := \mathbb{R}^2 \rtimes \mathrm{GL}^+(\mathbb{R}^2)$$

with the group operation given by

$$(x_2, A_2)(x_1, A_1) := (A_2 x_1 + x_2, A_2 A_1)$$

for all $(x_2, A_2), (x_1, A_1) \in \mathrm{Aff}^+(\mathbb{R}^2)$.

Where $\mathrm{GL}^+(\mathbb{R}^2)$ is the group of 2-dimensional real linear transformations with positive determinants, i.e. 2×2 matrices with positive determinant. This positive determinant ensures that the transformation does not flip the image, as the Gaussians and their derivatives are symmetric, which is not needed, and only confuses.

Todo

What are the unit element and the inverse?

What is the dimension of the group?

Prove $\mathrm{Aff}^+(\mathbb{R}^2)$ is a *Lie group*.

The affine group acts on $\mathbb{R}^2$ in the following manner:

$$g \triangleright y = (x, A) \triangleright y := Ay + x$$

for all $y \in \mathbb{R}^2$ and $g = (x, A) \in \mathrm{Aff}^+(\mathbb{R}^2)$. Similarly, for subsets $S \subset \mathbb{R}^2$, we define

$$g \triangleright S := \{g \triangleright y \mid y \in S\}.$$

This allows us to use this Lie group to position points in our image. The next thing to consider is that the Gaussians we use to fill our image are not one point, but a collection of points defined by a function in $\mathbb{R}^2$. Therefore, we will also need to define how this group acts on a function on $\mathbb{R}^2$. This is induced as follows:

Let $f : \mathbb{R}^2 \to \mathbb{R}$ then we define $g \triangleright f : \mathbb{R}^2 \to \mathbb{R}$ by

$$(g \triangleright f)(y) := f(g^{-1} \triangleright y).$$

2.2.2 Lie groups & parameter space

This works quite well for positioning the Gaussians in some particular spot. However, to make this match our target image, we will use gradient descent, as will be explained later. This requires the parameters to lie in a vector space. This is not the case by default for the Lie group. This means that when applying gradient descent at the current stage, the parameters could no longer be in the Lie group after gradient descent steps. This means that this parameter has no longer any useful meaning, and cannot be used to position the Gaussian.

For this reason, we will search for a parameter space, V, such that V is a vector space, and there exists a surjective function $\psi : V \to G$. We can use this to apply gradient descent on V, as this is a vector space, and render the image using G.

2.2.3 The Lie algebra $\mathfrak{aff}(\mathbb{R}^2)$

To solve this issue, we will make use of the Lie algebra. This is the tangent space at the identity of the Lie group. The Lie algebra can map to the Lie group (and vice versa), but is also a vector space.

The Lie algebra of $\mathrm{Aff}^+(\mathbb{R}^2)$ is $\mathfrak{aff}(\mathbb{R}^2) := \mathbb{R}^2 \rtimes \mathfrak{gl}(\mathbb{R}^2) \equiv \mathbb{R}^2 \rtimes \mathbb{R}^{2\times 2}$. Note that we say $\mathfrak{aff}$ and not $\mathfrak{aff}^+$; this is because $\mathfrak{aff}$ is the Lie algebra of both the Aff^+ and Aff Lie groups, illustrating that there is not a one-to-one relationship between Lie groups and Lie algebras.

The Lie group $\mathrm{Aff}^+(\mathbb{R}^2)$ and Lie algebra $\mathfrak{aff}(\mathbb{R}^2)$ both have a matrix representation, respectively these are

$$\begin{pmatrix} A & x \\ 0 & 1 \end{pmatrix} \text{ and } \begin{pmatrix} W & v \\ 0 & 0 \end{pmatrix}.$$

This representation of the Lie group still follows the group operation, by doing the matrix multiplication.

2.2.4 The exponential & logarithmic map

Lie groups and Lie algebras are related to each other through the exponential map. For our case the exponential map $\exp_{\mathrm{Aff}^+(\mathbb{R}^2)} : \mathfrak{aff}(\mathbb{R}^2) \to \mathrm{Aff}^+(\mathbb{R}^2)$ is given by exp.

Doing this for W and v seperately is equivalent to

$$\exp((v, W)) := \left(\left(\int_0^1 e^{tW} \, \mathrm{d}t \right) v, e^W \right). \tag{1}$$

Unfortunately, from (Culver 1966) we know that the Aff^+ exponential map is not surjective. So we cannot use $\mathfrak{aff}$ to create every possible Gaussian. Besides the fact that it is not surjective, is the integral also a problem. This means that the function is not a convenient closed formula, but instead requires a lot of computation.

2.3 Kaji-Ochiai parameterization

As we cannot use the default Lie algebra $\mathfrak{aff}$ from the Lie group Aff^+ as V, we will need to look a bit further. We adapt the findings from (Kaji and Ochiai 2016), where they describe a parameterization of $\mathrm{Aff}^+(\mathbb{R}^3)$, where this parameterization has a similar role as the Lie algebra. This parameterization is for a 3D affine transformation; it starts from what is essentially the Lie algebra but constructs a different surjective mapping than the exponential onto the group (but still closely related). We will simplify this parametrization to 2 dimensions.

The Kaij-Ochiai parameterization works by factoring the Lie algebra of $\mathrm{Aff}^+(\mathbb{R}^2)$ as per

$$\mathfrak{aff}(\mathbb{R}^2) \equiv \mathbb{R}^2 \times \mathfrak{so}(2) \times \mathfrak{sym}(2) \equiv \mathbb{R}^2 \times \mathbb{R} \times \mathbb{R}^3 \equiv \mathbb{R}^6$$

where $\mathbb{R}^2$ encodes the translation generator, $\mathfrak{so}(2)$ the rotation generator and $\mathfrak{sym}(2)$ the scale and shearing generator. Here $\mathfrak{so}(2)$ is the Lie algebra of $\mathrm{SO}(2)$ and is given by the real 2×2 anti-symmetric matrices, and so has only a single degree of freedom:

$$\mathfrak{so}(2) = \left\{ \begin{pmatrix} 0 & -r \\ r & 0 \end{pmatrix} \middle| \, r \in \mathbb{R} \right\}.$$

The second vector space $\mathfrak{sym}(2)$ is the set of (real) symmetric 2×2 matrices, but it is not the Lie algebra of any Lie group since the Lie bracket of two symmetric matrices is not necessarily a symmetric matrix; it is anti-symmetric in fact:

$$[A, B]^T = (AB - BA)^T = B^T A^T - A^T B^T = -(A^T B^T - B^T A^T) = -[A, B]$$

for all $A, B \in \mathfrak{sym}(n)$.

The vector space $\mathfrak{sym}(2)$ has three degrees of freedom:

$$\mathfrak{sym}(2) = \left\{ \begin{pmatrix} s_1 & s_3 \\ s_3 & s_2 \end{pmatrix} \middle| \, s_1, s_2, s_3 \in \mathbb{R} \right\}.$$

Definition 2.3.1 The parametrization map $\phi : \mathbb{R}^2 \times \mathbb{R} \times \mathbb{R}^3 \to \mathrm{Aff}^+(\mathbb{R}^2)$ is given by

$$\phi(x, r, s) := \left(x, \exp \begin{pmatrix} 0 & -r \\ r & 0 \end{pmatrix} \exp \begin{pmatrix} s_1 & s_3 \\ s_3 & s_2 \end{pmatrix} \right). \tag{2}$$

The matrix exponentials in the parametrization map 2 have the following closed formulae

$$\exp_{\mathfrak{sym}}(r) = \exp \begin{pmatrix} 0 & -r \\ r & 0 \end{pmatrix} = \begin{pmatrix} \cos r & -\sin r \\ \sin r & \cos r \end{pmatrix}$$

and

$$\exp_{\mathfrak{so}}(s_1, s_2, s_3) = \exp\begin{pmatrix} s_1 & s_3 \\ s_3 & s_2 \end{pmatrix} = e^{\frac{s_1+s_2}{2}} \begin{pmatrix} \cosh(a) + \frac{s_1-s_2}{2a}\sinh(a) & \frac{s_3}{a}\sinh(a) \\ \frac{s_3}{a}\sinh(a) & \cosh(a) + \frac{s_2-s_1}{2a}\sinh(a) \end{pmatrix}$$
$$= e^{\frac{s_1+s_2}{2}} \left(\cosh(a)\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \frac{\sinh(a)}{a}\begin{pmatrix} \frac{s_1-s_2}{2} & s_3 \\ s_3 & \frac{s_2-s_1}{2} \end{pmatrix} \right)$$

where $a := \frac{1}{2}\sqrt{(s_1 - s_2)^2 + 4s_3^2}$.

Note that $\det\left(\exp\begin{pmatrix} 0 & -r \\ r & 0 \end{pmatrix}\right) = 1$ and $\det\left(\exp\begin{pmatrix} s_1 & s_3 \\ s_3 & s_2 \end{pmatrix}\right) = e^{s_1+s_2} > 0$.

Furthermore, we can also look at the inverse of the parameterization.

Definition 2.3.2 A locally differentiable inverse $\overleftarrow{\phi} : \mathrm{Aff}^+(\mathbb{R}^2) \to \mathbb{R}^2 \times \mathfrak{so}(2) \times \mathfrak{sym}(2)$ is given by

$$\overleftarrow{\phi}(x, A) := \left(x, \log\left(A(A^TA)^{-\frac{1}{2}}\right), \log\left(\sqrt{A^TA}\right)\right). \tag{3}$$

This local inverse relies on the *polar decomposition* of a square matrix (Hall 2015, § 2.5). Every square matrix A admits a polar decomposition into an orthogonal matrix R and a positive semi-definite matrix S so that $A = RS$. If A is invertible, which it is in our case, then the polar decomposition is unique and S is positive definite. The matrices R and S are calculated as per

$$R = A(A^TA)^{-\frac{1}{2}} \quad \text{and} \quad S = (A^TA)^{\frac{1}{2}}.$$

Since A is invertible A^TA is symmetric positive definite and so has a unique square root, which is itself symmetric positive definite and hence uniquely invertible. Clearly $A = RS$ and S is symmetric positive definite, orthogonality of R follows from

$$R^TR = \left(A(A^TA)^{-\frac{1}{2}}\right)^T\left(A(A^TA)^{-\frac{1}{2}}\right) = (A^TA)^{-\frac{1}{2}}(A^TA)(A^TA)^{-\frac{1}{2}} = I.$$

Note

Some formulae for computing an inverse (need to verify):

$$\begin{aligned} s_1 + s_2 &= \log(\det(S)) = \log(\det(A)), \\ \cosh(a) &= 2e^{-\frac{s_1+s_2}{2}}\,\mathrm{trace}(S), \\ s_1 - s_2 &= e^{-\frac{s_1+s_2}{2}}\frac{a}{\sinh(a)}(S_{1,1} - S_{2,2}), \\ s_3 &= e^{-\frac{s_1+s_2}{2}}\frac{a}{\sinh(a)}S_{2,1}. \end{aligned}$$

2.4 A note on shearing

Shearing is the changing of the angle of the orthogonal base vectors of the space. No shearing with respect to $\{e_1, e_2\}$ implies that $Ae_1^TAe_2 = 0$. In our case $e_1 := (1, 0)^T$ and $e_2 := (0, 1)^T$. This shearing can have some interesting properties, so we will take a closer look to see if we want this or if we want to exclude this from the Gaussian transformations.

Dissecting the polar decomposition from earlier, we can see that $R \in \mathrm{SO}(2)$ does rotation, and $S \in \mathrm{SPD}(2)$ does scaling and this shearing.

We define S as

$$S = \begin{pmatrix} s_1 & s_3 \\ s_3 & s_2 \end{pmatrix}$$

.

Then we can see that s_3 generates shearing, as when $s_3 = 0$ then we have $(Se_1)^T(Se_2) = (s_1, 0)(0, s_2)^T = 0$. And vice versa, if $s_3 \neq 0$, then $(Se_1)^T(Se_2) = (s_1, s_3)(s_3, s_2)^T = s_1 s_3 + s_3 s_2 \neq 0$ as $s_1, s_2 > 0$, clearly introducing a shear.

We know that the stabilizer of ϕ_0 is $\mathrm{SO}(2)$. I.e.

$$\mathrm{Stab}_G(\phi_0) := \{g \in G \mid g \rhd \phi_0 = \phi_0\} \equiv \mathrm{SO}(2)$$

Furthermore, we can see that the effect of a shear on a Gaussian can also be generated by a rotation and a scale. I.e., we can remove s_3 and still generate all Gaussian configurations.

Theorem 2.4.1 (Shear generation)

For all $A \in \mathrm{GL}^+(\mathbb{R}^2)$, there exists a $\overline{R} \in \mathrm{SO}(2)$ and $D \in \mathrm{SPD}(2)$ which is also diagonal, i.e. $D = \begin{pmatrix} e^{s_1} & 0 \\ 0 & e^{s_2} \end{pmatrix}$, such that $A \rhd \phi_0 = \overline{R}D \rhd \phi_0$.

Proof:

We know that $A = RS$, then we can do an Eigendecomposition and decompose S as

$$S = Q\Lambda Q^T.$$

Where $\Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$ with $\lambda_{\{1,2\}}$ as the Eigenvalues from S. $Q \in O(2)$, however we can flip the Eigenvalues in the decomposition when $\det(Q) = -1$, therefore without loss of generality $Q \in \mathrm{SO}(2)$.

This means that we get the following.

$$A \rhd \phi_0 = RS \rhd \phi_0 = RQ\Lambda Q^T \rhd \phi_0 = RQ\Lambda \rhd (Q^T \rhd \phi_0) = RQ\Lambda \rhd \phi_0.$$

Now we can define $\overline{R} := RQ \in \mathrm{SO}(2)$ and $D := \Lambda$.

Thus $A \rhd \phi_0 = \overline{R}D \rhd \phi_0$. ■

Proposition 2.4.2 (Polar decomposition ordering)

Reversing rotational and scaling in the parametrization of Kaij-Ochiai will remove the geometric properties of s_3. I.e.

$$A(r, s_1, s_2, 0) = \exp\begin{pmatrix} 0 & -r \\ r & 0 \end{pmatrix} \exp\begin{pmatrix} s_1 & 0 \\ 0 & s_2 \end{pmatrix}$$

never contains no shearing in $\{e_1, e_2\}$.

While

$$B(r, s_1, s_2, 0) = \exp\begin{pmatrix} s_1 & 0 \\ 0 & s_2 \end{pmatrix} \exp\begin{pmatrix} 0 & -r \\ r & 0 \end{pmatrix}$$

can contain shearing in $\{e_1, e_2\}$.

Proof:

We will show that any transformation $A = RS$ with

$$R = \begin{pmatrix} \cos r & -\sin r \\ \sin r & \cos r \end{pmatrix}$$

and

$$S = \begin{pmatrix} \exp(s1) & 0 \\ 0 & \exp(s2) \end{pmatrix}$$

applied as RSx results in $e_1 e_2^T = 0$.

We can see this by

$$\begin{aligned} (Ae_1)^T(Ae_2) &= (RSe_1)^T(RSe_2) \\ &= e_1^T S^T R^T RSe_2 \\ &= e_1^T S^T Se_2 \qquad \text{As R is orthogonal} \\ &= (s_1, 0)^T(0, s_2) \\ &= 0. \end{aligned}$$

It is also true that for $B = SR$, applying SRx in the other order can result in shearing; we will show this with an example.

Define

$$S = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

and

$$R = \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}.$$

Now clearly $(SRe_1)^T(SRe_2) = -\frac{3}{2} \neq 0$

■

And thus we can see that the parametrization is still surjective with $s_3 = 0$. We can therefore remove this in the parameter space $V \equiv \mathbb{R}^5$. We can still map this using ψ, however, this will no longer map to Aff^+, but a subset of that where there exists no shearing. This subset is not a group.

Proposition 2.4.3

Define $T \subset \mathrm{GL}^+(\mathbb{R}^2)$. $T := \left\{A \in \mathrm{GL}^+(\mathbb{R}^2) \mid (Ae_1)^T Ae_2 = 0\right\}$. Then T is not a group.

Proof:

Take R and S, which are each clearly in this subset that contains no shearing. However, as we showed before, SR does contain shearing. Thus, not in this subset; therefore, the subset is not a group.

■

Intuitively, the previous results make sense as the rotational symmetry of ϕ_0 makes it such that the basis choice $\{e_1, e_2\}$ is arbitrary and has no dependency on anything else. For ϕ_1 and ϕ_2, this choice is relevant, as we already take the derivative in one of these directions. Shearing will add a unique property to these. Below is an example of the concept of shear in ϕ_0 being generated without s_3 or fully with s_3, and the effects this has on ϕ_1 and ϕ_2.

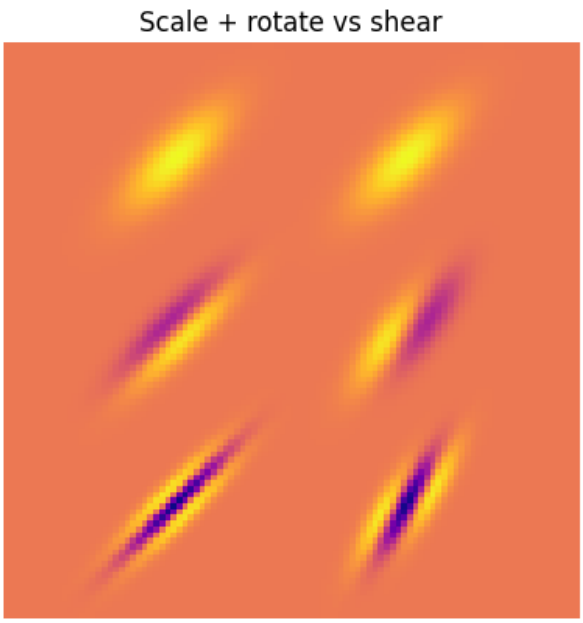


Figure 3: Left: Shearing via scale & rotate, $s_3 = 0$ | Right: Shearing via setting s_3

While this is not exactly the same, and for $\phi_{\{1,2\}}$ parameterization with $s_3 = 0$ is no longer surjective, however, removing this property would give us some better analysis on the anisotropy[1] of a Gaussian, as the anisotropy is now uniquely generated by the scaling parameters. However, we will need to do some experiments if the shear property has a significant impact on the first and second derivatives.

3 Methods

We will look at the methods that we will be using to actually create the Gaussians representations.

3.1 Rendering function

The target image we want to approximate is defined as $f(x) : \mathbb{R}^2 \to \mathbb{R}^3$.

The final image that is generated from the parameters is defined by

[1] Anisotropy is how "stretched" an ellipsoid, or in our case a Gaussian, is. We will see later why this is useful

$$\hat{f}(x) = \sum_{k=0}^{n} \sum_{i}^{3} c_k^i (g_k \triangleright \phi)(x) = \sum_{k=0}^{n} \sum_{i}^{3} c_k^i (\psi(v_k) \triangleright \phi)(x)$$

To find the optimal parameters c and v, we will use gradient descent as explained earlier. For this, we will define a score, called a loss, that determines how well a set of parameters represents the image. Taking steps downwards of the gradient will find us a local minimum. Using extra methods such as optimizers, scheduler, and modifying the loss, we will make sure that minimum found is as close to the global minimum as possible.

3.2 Culling

As we want to decrease the number of Gaussians that we save, we will remove the Gaussians based on their parameters. This process we will call culling. The most obvious is that we would like to remove Gaussians where the color values are negligibly small for all Gaussian layers. Formalized by

$$\sum_{j}^{3} \left|c^{1,j}\right| \leq 0.05 \wedge \sum_{j}^{3} \left|c^{2,j}\right| \leq 0.05 \wedge \sum_{j}^{3} \left|c^{3,j}\right| \leq 0.05.$$

The other criterion is again the anisotropy of the Gaussian, as mentioned earlier. Formally, we do this by removing all Gaussians where

$$s1 + s2 \leq -7.5.$$

The problem is that these Gaussians are visible, so doing this culling after all training is done would leave some gaps in the image. Therefore, we do this on 80% of the training, thus giving the gradient descent still some iterations to recover and fill the gaps.

4 Results

To put this into practice, we use pytorch to make use of CUDA kernels and speed up the rendering. The source code is available at (). The images are constrained to 1:1 to ease the implementation; the theory can be easily scaled to any ratio. The resolution can differ, depending on the input. For the experiments, mostly 100x100 resolutions are used, as this is a limitation of the physical hardware on which the tests were run.

The hardware used is an `Intel(R) Core(TM) i7-10750H (12) @ z` and `NVIDIA Quadro T1000 Mobile`; on this hardware, training took between 1 minute and 5 minutes. However, this is not a goal of the project.

In this section, we will discuss the results and adaptations we have made based on the findings. We will first present the final product, which includes all the adaptations, which we will use as our base performance to compare against.

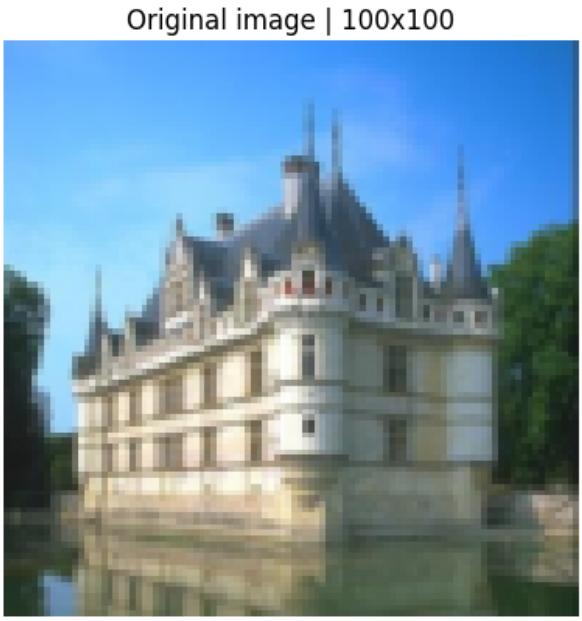


Figure 4: Original castle image

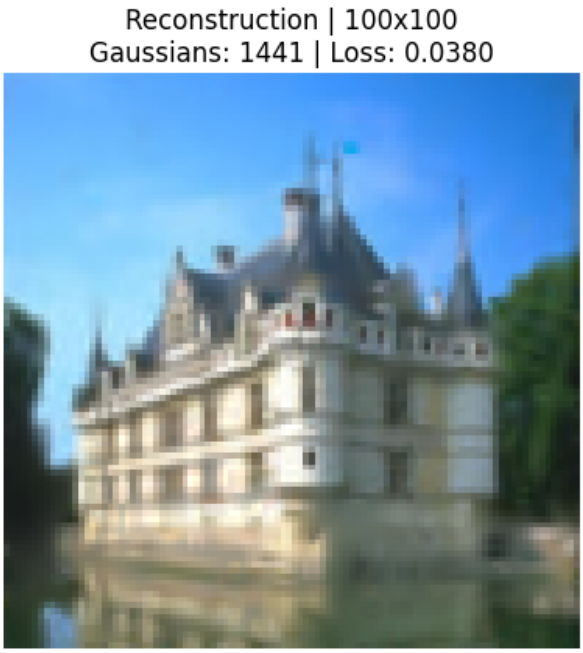


Figure 5: Base performance of the representation

4.1 Initalization

For the initialization, we use a hyperparameter for the number of Gaussians that will be spawned. The color parameters of ϕ_0 will be initialized with a normal distribution $N(0,1)$, for ϕ_1 and ϕ_2, this is $N(0,0.1)$. This prevents the first and second orders from being too active. The position of the Gaussians is also initialized with $N(0,1)$. The scaling parameters are also generated with $N(0,1)$ but immediately shifted down by -4. If this is not done, the Gaussians will overlap too much and fight against each other, resulting in streaks. This can be seen below.

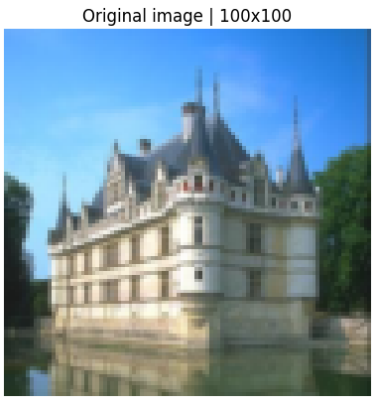


Figure 6: Original castle image

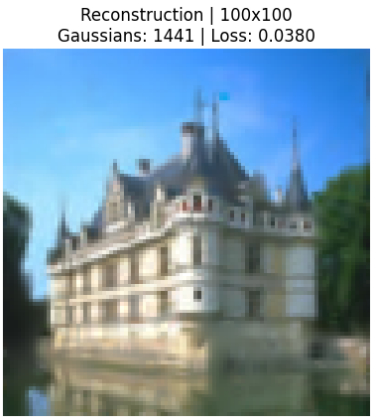


Figure 7: Base performance of the representation

Figure 8: Representation with unscaled initialization

4.2 Gradient descent

As mentioned earlier we use gradient descent. For this we need a loss function. This loss functions consists of a few different components. Most importantly, the L_1 loss, formally defined as $\|f - \hat{f}\|$. Besides this, we will also use the structural similarity index measure (SSIM). These will be weighed with λ. The total loss for how well the image looks will be

$L_{\text{image}} = \lambda \|f - \hat{f}\| + (1 - \lambda) * \left(1 - \text{SSIM}\left(f, \hat{f}\right)\right)$. The L_1 loss has a big focus on individual pixel differences, while this is useful, this might overfit on certain pixel. SSIM on the otherhand focus on differences on a structural level, looking more at luminance and constrast. Combining these two has been done in similiar image representation projects, and work here again.

> **Todo**
>
> Possible: add some images to back this up. Perhaps also that the loss is more "convex" (i.e. doesnt go up, which was the case when only using L1 loss). Although I have no good reasoning for why this happens.

Besides the actual representation, we have two other things that we want to discourage. As gradient descent tries to find the minimum of the loss, we can put things in the loss to discourage certain behavior. First of all, we want to limit the anisotropy, to make sure Gaussians do not become stretched out, but stay fairly round. This will prevent artifacts and overfitting. Furthermore, we also do not want Gaussians to become too small; to be exact, they must not be smaller than a pixel, as this will make them hidden while rendering, but they still contain information in the final embedding.

The effects of small gaussians is not directly apparent, contrary, it seems at some points the representation is better. Noticing the difference can be done by looking at the detail in the dormer window. The difference in the amount of Gaussians will be explained in "Culling".

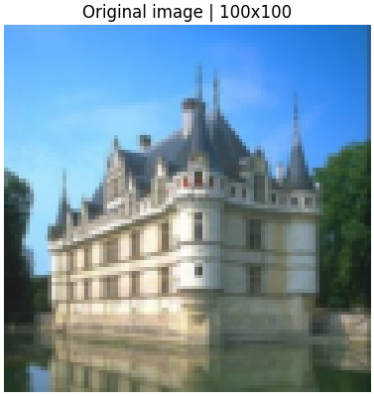


Figure 9: Original castle image

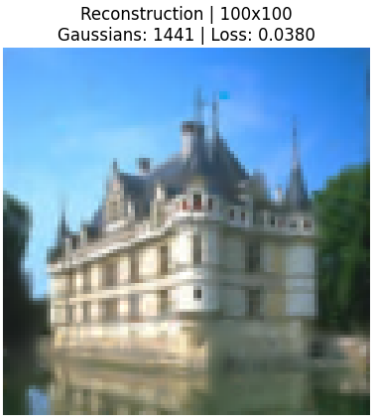


Figure 10: Base performance of the representation

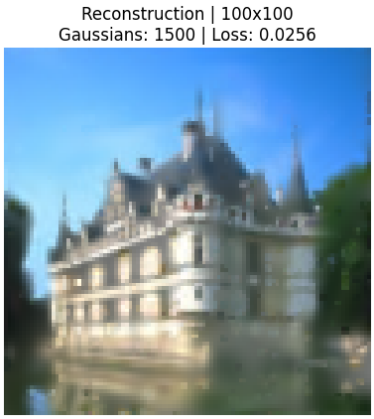


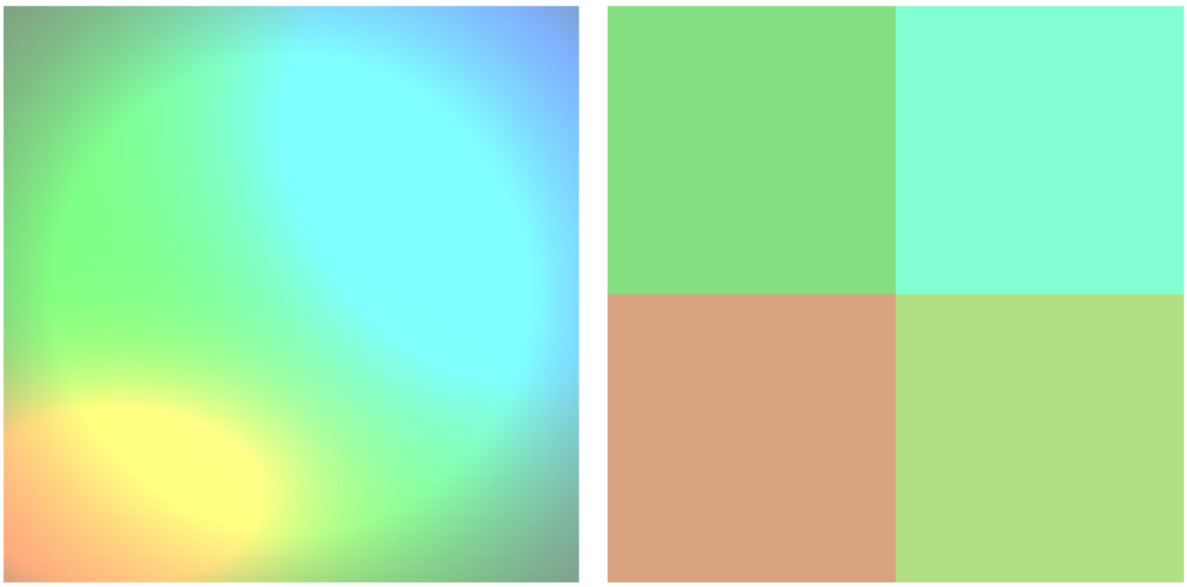
Figure 11: No sizing in loss

However when rendering the representations upscaled, the difference becomes apparant. Both images are trained in a 100x100 resolution, but the final render is upscaled to 150x150.

Figure 12: Base performance of the representation, upscaled

Figure 13: No sizing in loss, upscaled

You will see the artifacts in the right image. These are not visible in the 100x100, as they are smaller than a pixel in the 100x100 render. This comes from the fact that the representation is infinitely detailed, but the rendering is a sampling of that representation. For each pixel, instead of the average over the whole pixel, only a specific spot within the pixel is picked. Taking the average is computationally expensive, and only shifts the same problem to the smaller level. These small Gaussians do influence the sample spots of the pixels, so removing them is not an option.



100x100 2x2

Figure 14: The same Gaussian representation, sampled differently. In this implementation, the sampling happens at the bottom left corner. The bright yellow spot is not visible anymore.

To already penalize this in the trainig process, we will add the following loss function $L_{\text{sizing}} = \overline{\exp(-(s_1 + s_2) - 8)} * \lambda_{\text{sizing}}$ to the loss. Where λ_{sizing} determines how much influence this loss functions has.

The effect of anisotropic Gaussians is fairly similiar, as can be seen below.

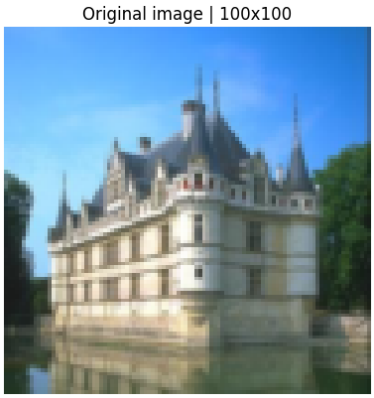


Figure 15: Original castle image

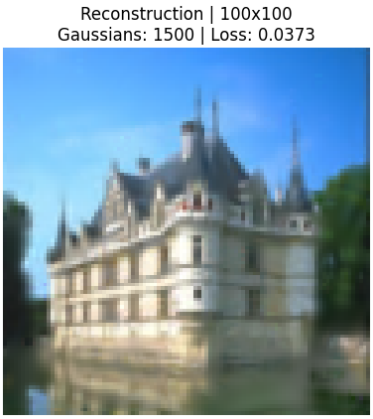


Figure 16: Representation without shearing loss

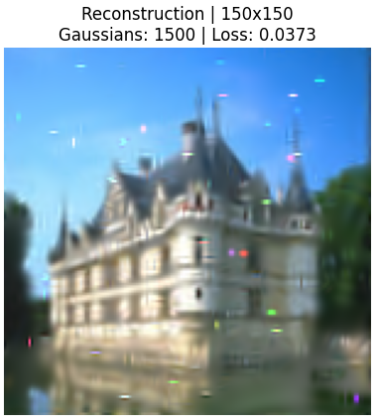


Figure 17: Representation without shearing loss, upscaled

Here we define the anisotropy as $L_{\text{anisotropy}} = \overline{|s_1 + s_2|} * \lambda_{\text{anisotropy}}$. Because shear is exclusively generated by the size, this is an easy way to penalize for this.

The final loss is then defined as $L = L_{\text{image}} + L_{\text{anisotropy}} + L_{\text{sizing}}$

As $\mathbb{R}^2 \times \mathbb{R} \times \mathbb{R}^3$ are all vector spaces, and the loss function is continuous, we can use gradient descent to optimize this. We will use AdamW and a scheduler to further optimize the training process.

4.3 Amount of Gaussians

The amount of Gaussians have a significant impact on both the quality of the image and the amount of tokens, but they are inversely related. Finding an optimal is therefore essential. This amount is also highly dependent on the resolution, and nature of the image. Images which a lot of contrast and high frequency data, such as text, require more Gaussians.

The base image we have been using uses 1500 Gaussians, below are the effects doubling and halving that

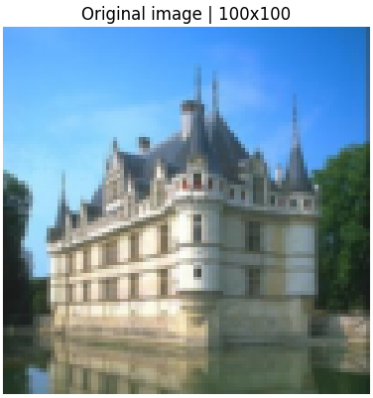


Figure 18: Original castle image

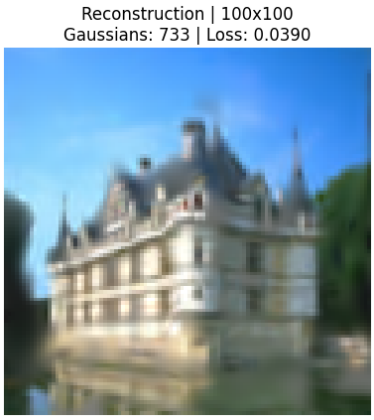


Figure 19: Representation with 750 Gaussians initialized

SSIM Loss: 0.0770
L1 Loss: 0.0186

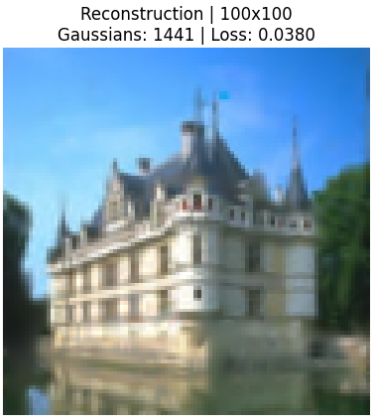


Figure 20: Representation with 1500 Gaussians initialized

SSIM Loss: 0.0429
L1 Loss: 0.0131

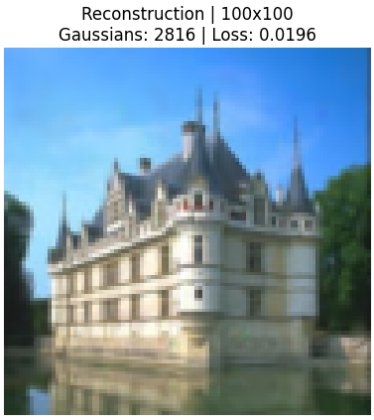


Figure 21: Representation with 3000 Gaussians initialized

SSIM Loss: 0.0088
L1 Loss: 0.0056

Clearly the progression is visible, the representation initalized with 3000 Gaussians is almost identical for the naked eye. An important side not here is that the tokens grow exponentionally in terms of cost of the neural network.

4.4 Higher order Gaussians

To see the performance and the influence of the higher order Gaussians, we remove the ϕ_0 and only render $\phi_{\{1,2\}}$. Looking at the previous comparison we can clearly see the lines and edges in the image.

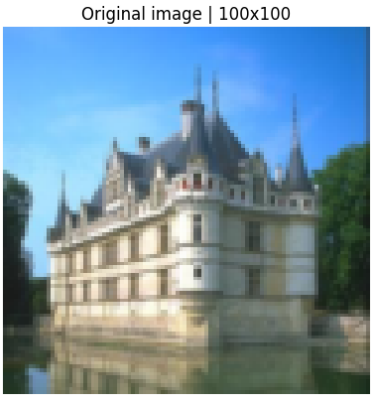


Figure 22: Original castle image

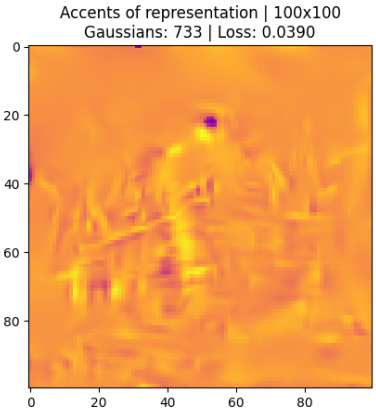


Figure 23: Accents with 750 Gaussians initialized

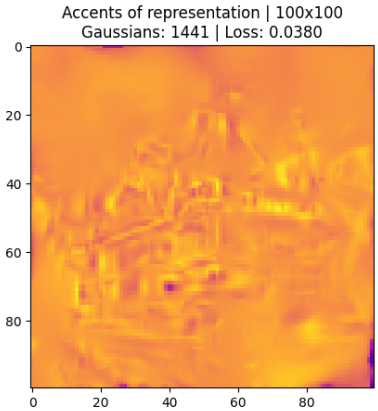


Figure 24: Accents with 1500 Gaussians initialized

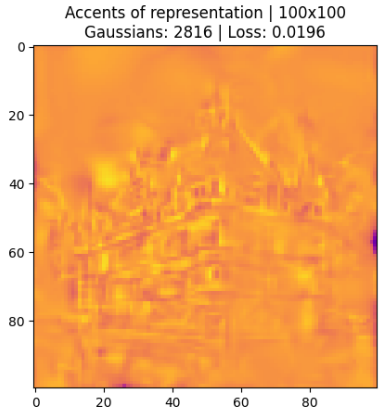


Figure 25: Accents with 3000 Gaussians initialized

You can clearly see where the image has higher frequency, and see the outlines of the castle. This effect is a lot stronger with more Gaussians.

Figure 26: Original castle image

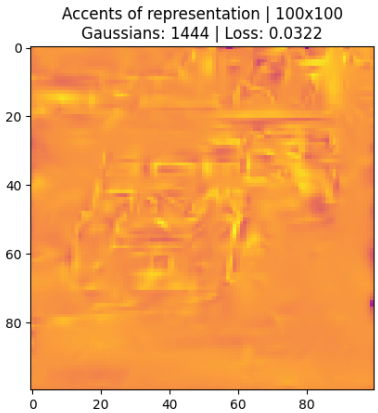


Figure 27: Accents with 1500 Gaussians initialized

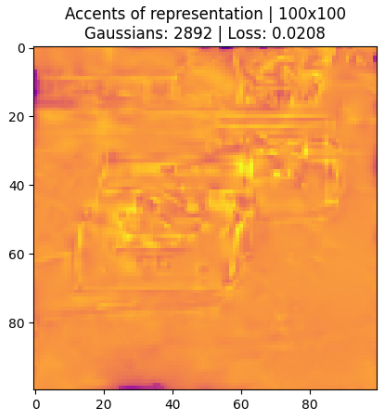


Figure 28: Accents with 3000 Gaussians initialized

On the cars image containing more edges, this effect is more clearly visible.

5 Appendix section

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